

WhatsApp AI Assistant - Knowledge Base

About This AI Assistant

This is an AI-powered WhatsApp chatbot that uses Retrieval-Augmented Generation (RAG) to provide accurate and contextual responses. The assistant can answer questions based on the documents in its knowledge base, as well as use its general knowledge for other queries.

Features

The WhatsApp AI assistant offers several key features: 1. Natural language understanding through advanced language models 2. Document-based question answering using RAG technology 3. Conversation history to maintain context across messages 4. Real-time responses through WhatsApp integration 5. Secure and private communication

How to Use

To interact with the AI assistant: 1. Send a message to the configured WhatsApp number 2. Ask questions about the topics in the knowledge base 3. The AI will retrieve relevant information and provide answers 4. You can have multi-turn conversations 5. The assistant remembers context from recent messages

Technology Stack

This assistant is built using: - Python for backend development - Streamlit for the user interface - Twilio API for WhatsApp integration - Hugging Face models for language understanding - ChromaDB for vector storage and semantic search - LangChain for orchestrating the RAG pipeline

RAG (Retrieval-Augmented Generation)

RAG is a technique that enhances language models by: 1. Breaking documents into smaller chunks 2. Converting chunks into vector embeddings 3. Storing embeddings in a vector database 4. Retrieving relevant chunks for each query 5. Using retrieved context to generate accurate responses This approach ensures the AI provides factual, document-grounded answers rather than relying solely on its training data.

Privacy and Security

Your privacy is important: - Conversations are processed securely - No data is stored permanently on external servers - The system can be deployed locally for complete control - You can delete conversation history at any time - Documents remain in your local environment